

A Universal Quartic Dispersion Framework: Numerical Validation, Multi-Messenger Time-of-Flight Bounds, and Condensed-Matter Analog Mappings

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Abstract

We consolidate and numerically validate the one-parameter quartic dispersion framework $\omega(k) = ck\sqrt{1 + \Lambda k^2}$ introduced in Papers 1–2, and extend it in three directions. **(1) Numerical infrastructure:** we implement and validate a spectral solver for the corresponding wave equation, verify the discrete biharmonic operator D_n^4 converges to the continuum limit at the expected $O(h^2)$ order, and demonstrate that Λ can be recovered from noisy (k, ω) data to sub-percent accuracy. **(2) Corrected Fermi-LAT audit:** we identify and correct a factor-of-3/2 normalization error in the group-velocity relation used in Paper 1’s Fermi-LAT bound, tightening it from $\Lambda < 2.16 \times 10^{-53} \text{ m}^2$ to the audited value $\Lambda < 1.44 \times 10^{-53} \text{ m}^2$. **(3) Condensed-matter analog mappings:** we show that the same dispersion form appears in Bogoliubov excitations of Bose–Einstein condensates ($\Lambda = \xi^2/4$), an analytically verified mapping, and we release open-source fitting tools so that this claim is directly falsifiable by experimentalists. We also examined a proposed mapping to trigonal-warping-corrected Dirac materials and one to photonic k -space dispersion; on closer inspection neither currently rests on a literature-supported derivation (Section 5), and we report this honestly rather than presenting them as validated. We emphasize throughout the distinction between an *audited observational bound* (Fermi-LAT), a *validated theoretical sensitivity ceiling* (EHT), and an *illustrative forecast* (gravitational-wave multi-band observations) — only the first constitutes a genuine fitted constraint from published data.

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1 Introduction

Papers 1 and 2 established a one-parameter Hamiltonian dispersion framework,

$$H = \frac{1}{2} [g^{\mu\nu} k_\mu k_\nu + \Lambda k_{\text{loc}}^4], \quad (1)$$

and derived its consequences in Schwarzschild and Kerr spacetime: two proved theorems on field/worldline non-equivalence (Paper 1), and four results on photon-ring chromaticity, frequency-dependent shadow sensitivity, superradiance invariance, and eikonal quasinormal modes (Paper 2).

The present paper serves three purposes:

1. **Close the numerical-infrastructure gap.** Papers 1–2 used custom root-finding and implicit-differentiation code for each specific calculation (photon rings, QNM eikonal parameters). Here we build and validate a general-purpose PDE solver for the underlying wave equation, independent of any particular spacetime, and confirm it reproduces the known dispersion relation to controlled numerical accuracy.
2. **Audit and correct the Fermi-LAT bound.** A normalization error in the group-velocity relation used in Paper 1 is identified and corrected here (Section 3).
3. **Extend the physical scope.** We show that the flat-space limit of the dispersion relation, $\omega = ck\sqrt{1 + \Lambda k^2}$, is mathematically identical to the established Bogoliubov dispersion law for BEC excitations, and we release open-source fitting tools so that this claim is directly falsifiable by experimentalists. We also audit two further proposed condensed-matter mappings (Dirac materials, photonic crystals) and report that neither currently survives scrutiny in its present form (Section 5).

Remark 1. *This paper does not claim a unified physical origin for Λ across gravitational and condensed-matter contexts. It claims only that the same functional form of dispersion relation appears in each context, with different physical identifications of the parameter. A single numerical value of Λ that is simultaneously consistent across all probes would be evidence for common origin; this has not been established and is not claimed here.*

2 Numerical Wave-Equation Solver

2.1 Flat-space wave equation

The flat-space limit of the Hamiltonian (with ZAMO/comoving observer reducing to the ordinary observer) gives the real-space PDE

$$\partial_t^2 \phi = c^2 \nabla^2 \phi - c^2 \Lambda \nabla^4 \phi, \quad (2)$$

i.e. the standard wave equation with a biharmonic correction. We implement a 2D periodic-domain Fourier spectral solver (`wave_equation_2D_solver.py`) in which the spatial operator is exact (machine precision) and the only numerical error comes from finite time-sampling in the frequency-extraction step.

Table 1: Numerical vs. exact dispersion relation, $k = 5$.

Λ	ω_{exact}	$\omega_{\text{numerical}}$	relative error
0.00	5.00000000	4.98333333	3.33×10^{-3}
0.01	5.59016994	5.57153604	3.33×10^{-3}
0.05	7.50000000	7.47500000	3.33×10^{-3}
0.10	9.35414347	9.32296299	3.33×10^{-3}

2.2 Validation: recovered dispersion relation

Table 1 shows the numerically extracted $\omega(k)$ for a plane-wave initial condition, compared to the exact formula, at several Λ values (grid: 64×64 , wavevector $(k_x, k_y) = (3, 4)$, so $k = 5$).

The residual 3.33×10^{-3} relative error is uniform across Λ and is entirely attributable to FFT frequency-bin discretization in the time domain (finite number of oscillation periods sampled); it is not a spatial discretization error, since the spectral spatial operator is exact. Section 2.4 confirms this error scales as $O(1/N_{\text{samples}})$.

2.3 Discrete biharmonic operator D_n^4

For finite-difference (as opposed to spectral) implementations, the biharmonic term in Eq. (2) is approximated by $D_n^4 = (D_n^2)^2$, the composition of the standard 3-point discrete Laplacian. We verify (`paper3_Dn4_test.py`):

- ✓ **Symbol convergence:** the discrete Fourier symbol $[-(2/h^2)(1 - \cos kh)]^2$ converges to the continuum k^4 at measured order 2.000 (5 grid refinements, $h \in [0.01, 0.5]$), matching the expected order for a standard 5-point stencil.
- ✓ **Eigenfunction check:** direct matrix application of D_n^4 to a discretized plane wave reproduces the discrete-symbol prediction to 2×10^{-10} (machine precision), confirming the matrix construction is correct independent of the continuum-limit question.

2.4 Grid and temporal convergence

Table 2 confirms $O(1/N)$ convergence of the frequency-extraction error with the number of time samples, consistent with standard FFT binning behavior and ruling out spurious numerical drift in the solver.

Table 2: Temporal resolution convergence ($\Lambda = 0.05$, fixed spatial grid).

N_{samples}	relative error	convergence order
100	1.00×10^{-2}	—
300	3.33×10^{-3}	−1.000
1000	1.00×10^{-3}	−1.000
3000	3.33×10^{-4}	−1.000
10000	1.00×10^{-4}	−1.000

2.5 Λ recovery from noisy data

We further test the inverse problem directly relevant to experimentalists: given noisy (k, ω) pairs, can Λ be recovered by least-squares fitting? Using synthetic data with $\Lambda_{\text{true}} = 0.07$ and 0.1% Gaussian noise on ω , the fit recovers $\Lambda_{\text{fit}} = 0.069771$, a relative error of 0.33% — well within the noise level, confirming the fitting procedure introduces no significant bias.

3 Corrected Fermi-LAT Bound

3.1 The normalization error

Paper 1 derived a bound on Λ from Fermi-LAT gamma-ray-burst timing [Abdo et al., 2009] by equating the fractional group-velocity deviation $(v_g - c)/c$ to the standard linear-in- Λ term of the dispersion relation. Auditing this derivation reveals that the correct relation is

$$\frac{v_g - c}{c} = \frac{3}{2}(D - 1), \quad (3)$$

where $D = v_g/c$ was (incorrectly) treated as directly proportional to $\Lambda\omega^2$ without the leading numerical factor $3/2$ that arises from differentiating $\omega = ck\sqrt{1 + \Lambda k^2}$ to obtain the group velocity (Paper 1, Eq. for v_g ; the factor originates from $d/dk[k\sqrt{1 + \Lambda k^2}]$ at leading order in Λk^2).

3.2 Corrected bound

Result 1 (Audited Fermi-LAT bound). *Correcting for the $3/2$ factor gives*

$$\Lambda_{\text{corrected}} = \frac{2}{3} \Lambda_{\text{original}}, \quad (4)$$

i.e. a 33.3% tightening of the bound:

$$\boxed{\Lambda_{\text{Fermi}} = 1.4421 \times 10^{-53} \text{ m}^2} \quad (5)$$

in place of the previously quoted $2.1632 \times 10^{-53} \text{ m}^2$. The order of magnitude, $\Lambda_{\text{Fermi}} = O(10^{-53} \text{ m}^2)$, is unchanged.

This correction does not affect any result in Paper 2 (Kerr photon rings, chromatic shadow, superradiance, eikonal QNMs), which do not depend on the Fermi-LAT normalization.

4 Consolidated Multi-Probe Summary

Table 3 consolidates the three observational/theoretical probes of Λ developed across Papers 1–3, with an explicit statement of evidentiary status for each.

Remark 2 (Status definitions — read carefully). **Audited bound (Fermi-LAT):** *derived from a published, fitted constraint on Lorentz-invariance-violating time-of-flight dispersion [Abdo et al., 2009], with the Λ -model normalization independently verified in Section 3. This is a genuine observational constraint.*

Table 3: Consolidated Λ probes. See remarks below for status definitions.

Probe	Observable	Scaling	Result	Status
Fermi-LAT	time-of-flight	$\Lambda\omega^2$	$1.44 \times 10^{-53} \text{ m}^2$	audited bound
EHT M87*	chromatic shadow shift	$\Lambda\omega^2$	$2.6 \times 10^{-9} \text{ m}^2$	validated sensitivity
GW (LIGO/LISA)	propagation delay	$\Lambda\omega^2 K(z)$	$3.2 \times 10^{-10} \text{ m}^2$	forecast

Validated sensitivity ceiling (EHT): the value of Λ corresponding to a 10% shadow-size tolerance model calibrated to published EHT angular resolution at 230 GHz (Paper 2, Section 5). This is not a fit to actual M87* shadow data; it is the sensitivity that such a fit could in principle achieve under stated assumptions. A genuine EHT constraint requires per-band sensitivity, opacity, and imaging-systematics modeling not attempted here.

Forecast (GW): an illustrative projection of the sensitivity achievable from a hypothetical multi-band gravitational-wave detection with electromagnetic counterpart, using the FLRW propagation formula $\Delta T(\omega, z) = -(\Lambda/2c^3)\omega^2 K(z)$ derived in Paper 2’s FLRW extension. This is not a fitted LIGO/Virgo/KAGRA constraint; no such analysis has been performed on real GW170817 or other event data.

All three probes share the identical $\Lambda\omega^2$ scaling, which follows from the same analytical proof (vanishing of $g^{\mu\nu}k_\mu k_\nu$ at the relevant null trajectory) established independently in each spacetime (Schwarzschild for Fermi-LAT and GW forecast, Kerr for EHT).

5 Condensed-Matter Analog Mappings

The flat-space, zero-curvature limit of the dispersion relation is

$$\omega(k) = ck\sqrt{1 + \Lambda k^2}. \quad (6)$$

Of the three condensed-matter systems considered during this audit, only one currently has a literature-supported derivation of this exact form. We report all three honestly in Table 4, including the two that do not (yet) survive scrutiny, rather than omitting the negative results.

Table 4: Condensed-matter systems examined for the dispersion form Eq. (6). Status reflects the audit in this section, not the convenience of the algebra.

System	Λ mapping	Physical inputs	Status
BEC (Bogoliubov)	$\xi^2/4$	healing length ξ	established
Dirac/Weyl material	$(1 - \eta^2)v_F^2/4$	warping η, v_F	speculative, unsupported
Photonic (k -space)	β_4^k/c^2	coeff. β_4^k of k^4 in $\omega^2(k)$	not yet measurable

5.1 Bose–Einstein condensates

The Bogoliubov dispersion relation for excitations in a weakly interacting BEC, $\omega(k) = c_s k \sqrt{1 + (\xi k/2)^2}$, matches Eq. (6) exactly with $\Lambda = \xi^2/4$, verified directly by matching the k^4 coefficients of the two forms ($\omega^2 = c_s^2 k^2 + (\hbar^2/4m^2)k^4$ vs. $\omega^2 = c_s^2 k^2(1 + \Lambda k^2)$). This is an

analytic identity, not a numerical coincidence. `examples/example_BEC.py` confirms the fitting code correctly recovers $\Lambda = \xi^2/4$ from noisy synthetic Bragg-spectroscopy-like data (healing length $\xi = 5 \times 10^{-7}$ m, 2% noise, recovered to 5.4%) — this validates the *fitting procedure*; the physical mapping itself is independently established by the coefficient-matching argument above, not by the synthetic-data test.

5.2 Dirac and Weyl materials — audited and not supported

An earlier draft of this section proposed $\Lambda = (1 - \eta^2)v_F^2/4$ for trigonal/hexagonal-warping-corrected Dirac dispersion, citing Fu [2009]. On inspection, that citation does not support this formula. Fu [2009] derives a hexagonal warping term that is *cubic* in momentum,

$$H = v_F(k_x\sigma_y - k_y\sigma_x) + \frac{\lambda_w}{2}(k_+^3 + k_-^3)\sigma_z, \quad (7)$$

explicitly described by the author as the counterpart of cubic Dresselhaus spin-orbit coupling. Squaring to obtain the energy dispersion gives

$$E(k)^2 = v_F^2 k^2 + \lambda_w^2 k^6 \cos^2(3\theta), \quad (8)$$

an *anisotropic* k^6 correction, not the isotropic k^4 correction required by Eq. (6). No power of the warping strength or angular averaging converts a k^6 term into a k^4 term; these are structurally different corrections to the dispersion relation. We were unable to locate any literature derivation of an isotropic k^4 (Lambda-model) correction to Dirac dispersion from trigonal warping, tilt, or any other known mechanism.

Remark 3 (On the synthetic-data test). `examples/example_dirac.py` generates synthetic “ARPES-like” data directly from the formula $\Lambda = (1 - \eta^2)v_F^2/4$ and then fits that same formula to it. Such a test is guaranteed to succeed regardless of whether the underlying physical mapping is real; it demonstrates only that the fitting code is unbiased, and must not be read as evidence for the mapping. We flag this explicitly because an earlier version of this manuscript did present the resulting agreement (“0.67%”) as if it were evidence for the physics. It was not.

We retain $\Lambda = (1 - \eta^2)v_F^2/4$ in the released code, clearly marked as speculative (it raises a runtime warning on use), for exploratory purposes only. It should not be treated as a prediction of this framework, and Table 4 reports it as such.

5.3 Photonic dispersion — corrected derivation, not yet connected to a measurable quantity

An earlier draft of this section claimed $\Lambda = \beta_4^k/(12c)$, derived from an expansion $\omega(k) = ck + (\beta_4^k/24)k^4 + \dots$. That expansion is incorrect. Direct Taylor expansion of Eq. (6) gives

$$\omega(k) = ck + \frac{1}{2}\Lambda c k^3 + O(k^5), \quad (9)$$

i.e. the leading correction to $\omega(k)$ itself is *cubic* in k ; there is no k^4 term in $\omega(k)$ at any order coming from this model. The quartic structure appears cleanly only in $\omega^2(k)$, which is exact and closed-form:

$$\omega(k)^2 = c^2 k^2 + c^2 \Lambda k^4 \implies \Lambda = \beta_4^k/c^2, \quad \beta_4^k \equiv c^2 \Lambda, \quad (10)$$

where β_4^k is defined as the coefficient of k^4 in $\omega^2(k)$ (units m^4/s^2). This corrected formula is used throughout the released code.

Even with the corrected algebra, β_4^k is not yet connected to any standard, independently measurable photonics quantity. Standard telecom dispersion coefficients ($\beta_2, \beta_3, \beta_4 = d^n\beta/d\omega^n$) are Taylor coefficients of the propagation constant expanded around a nonzero pump/operating frequency — a structurally different expansion from the near- $k = 0$ form used here. We checked the closest real physical analogue, quartic dispersion in fiber-optical event-horizon experiments [Webb et al., 2014]: even in the idealized “pure-quartic” operating point ($\beta_2 = \beta_3 = 0$), the co-moving-frame frequency near the group-velocity-matched wavenumber behaves as a quartic *minimum* with no linear term at all, $\Omega(\Delta k) \propto -\beta_4\Delta k^4/(24\beta_1^5)$ — a different mathematical structure from the deformed-light-cone form of Eq. (6), which always retains a leading linear branch $\omega \rightarrow ck$. We do not currently have a derivation connecting measured fiber β_n coefficients to β_4^k as used here.

Remark 4 (On the synthetic-data test). *As in Section 5.2, `examples/example_photonic.py` fits synthetic data generated from the same formula it tests, which validates only the fitting code (now correctly recovering the corrected $\Lambda = \beta_4^k/c^2$), not any physical claim about real photonic media. We report this explicitly rather than describing it as experimental validation.*

6 Open-Source Tools

All code described in this paper is released under the MIT license in the `Lambda-model` repository, organized as:

```
Lambda-model/
  paper3/
    wave_equation_2D_solver.py      (Section 2)
    paper3_Dn4_test.py              (Section 2.3)
    paper3_grid_convergence.py      (Section 2.4-2.5)
    lambda_experimental_validator.py (fitting tool, all domains)
  examples/
    example_BEC.py
    example_dirac.py
    example_photonic.py
```

The validator (`lambda_experimental_validator.py`) accepts either raw (k, ω) CSV data or domain-specific physical parameters, and reports: the fitted Λ with standard error, the fit quality (R^2), a significance test against $\Lambda = 0$, and (for domain examples) a direct comparison to the theoretical mapping. This is intended as a turnkey tool for experimentalists to test their own data against the framework without needing to reimplement the fitting procedure. The BEC mapping (`lambda_from_BEC`) is presented as established; the Dirac mapping (`lambda_from_dirac`) raises a runtime warning identifying it as speculative and unsupported by the cited literature (Section 5.2); the photonic mapping (`lambda_from_photonic`) uses the corrected β_4^k/c^2 form but is explicitly documented as not yet connected to a measurable laboratory quantity. Domain example scripts print an explicit reminder that synthetic-data agreement validates the fitting code, not the physical mapping, wherever that mapping is not independently established.

7 Conclusions

We have consolidated the quartic-dispersion framework of Papers 1–2 with three additions:

1. A validated numerical solver and discrete biharmonic operator, with confirmed $O(h^2)$ spatial and $O(1/N)$ temporal convergence, and demonstrated Λ -recovery from noisy data to sub-percent accuracy.
2. A corrected Fermi-LAT bound, $\Lambda_{\text{Fermi}} = 1.4421 \times 10^{-53} \text{ m}^2$, tightened by 33% from the original Paper 1 value due to a group-velocity normalization error.
3. One analytically established condensed-matter analog mapping (BEC, $\Lambda = \xi^2/4$), with the fitting procedure confirmed unbiased on synthetic data; and an honest audit of two further proposed mappings (Dirac materials, photonic crystals), neither of which currently survives scrutiny as a validated physical claim (Section 5). All fitting tools are released as open source regardless of a given mapping’s status, so experimentalists can test any of them directly against real data.

The overall evidentiary hierarchy is now:

theory (proved) $\rightarrow$ numerical validation (this paper) $\rightarrow$ audited bound (Fermi-LAT) $\rightarrow$ validated sensitivity ceiling (EHT) $\rightarrow$ forecast (GW) $\rightarrow$ open experimental test (condensed matter).

We emphasize that no single value of Λ has been simultaneously fitted across all probes; the framework remains a falsifiable hypothesis about a shared mathematical structure, not a confirmed physical law. The open-source tools released here are intended to enable that falsification test directly, by domain experts, on real data.

References

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